BITS Pilani
K K Birla Goa Campus

8086/8088 Hardware Specifications

MICROPROCESSORS & INTERFACING (CS F241)

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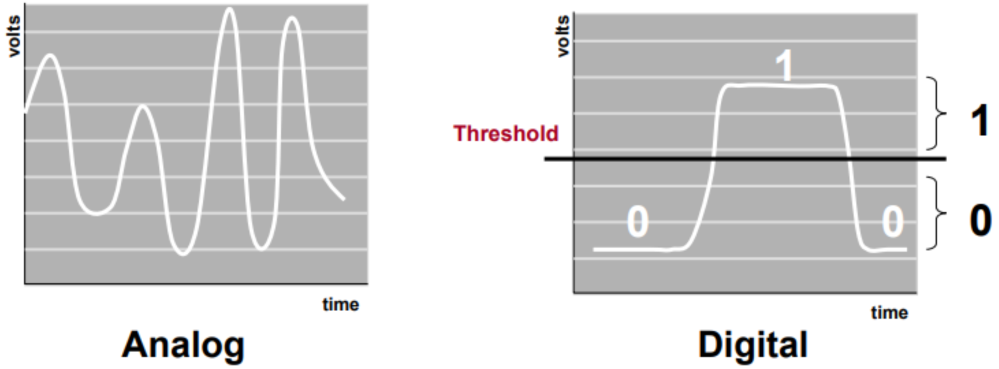
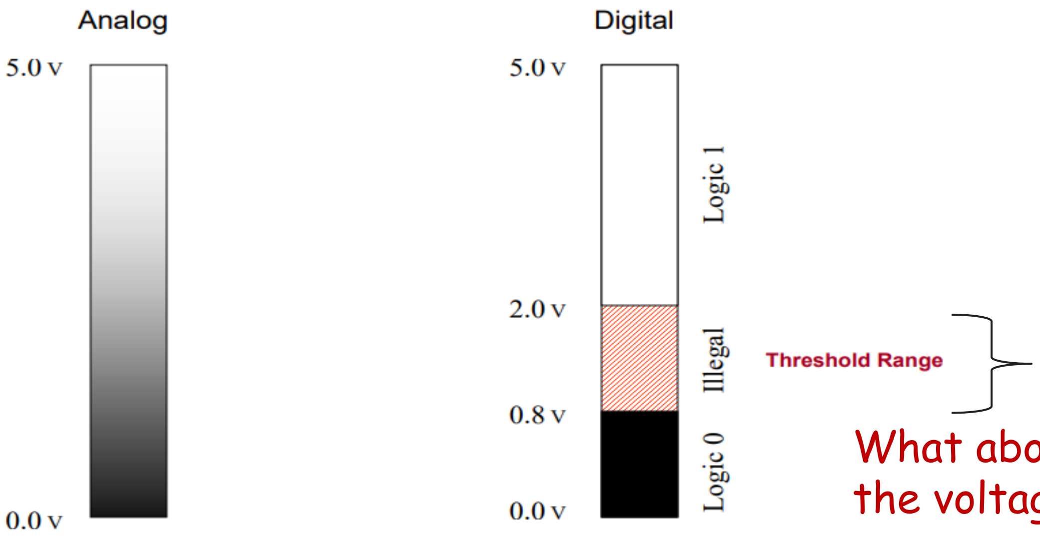
Department of CS & IS

Signal Types



- Digital signals are **just voltages**…
- Analog signal – Continuous time signal where each voltage level has a unique meaning
- Digital signal – Continuous signal where voltage levels are mapped into 2 ranges meaning 0 or 1

Analog and Digital Signal Meaning



What about
the voltage values here?

Each voltage value
has unique meaning

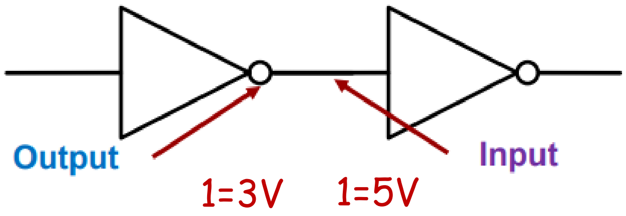
Each voltage value
Is mapped to ‘0’ or ‘1’

Noise Margin

How much noise the logic gate can tolerate without affecting its normal operation.

Importance: 1. Compatibility of circuit elements/gates.
2. Circuit performance stability.

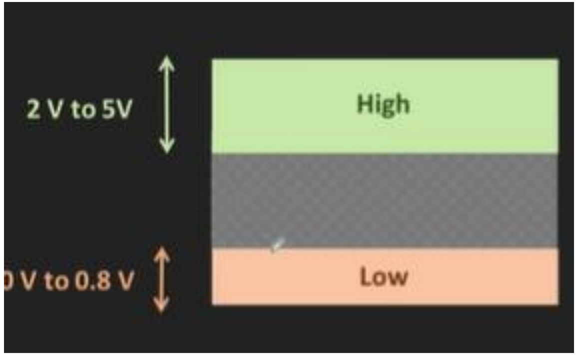
Noise Margins

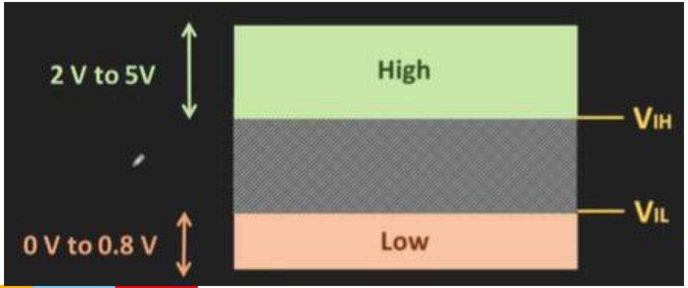


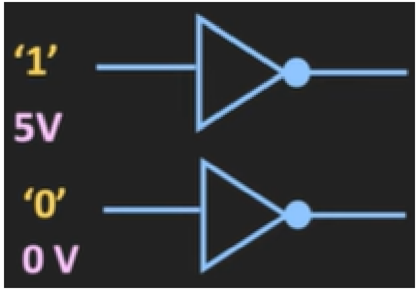
Consider the output of one digital circuit feeding the input of another

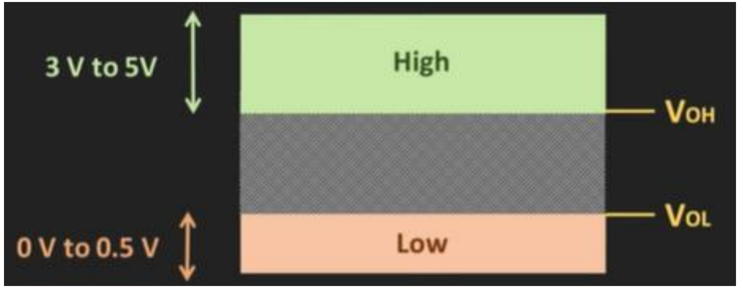
– Assume the devices are from different vendors (families of devices)

There may be different limits and requirements of the two devices – Example: The output may produce 3V to mean logic '1' while the next device's input requires 5V to be used as logic ‘1’


2 V to 5V
High
0 V to 0.8 V
Low


2 V to 5V
High
VIH
VIL
0 V to 0.8 V
Low


'1'
5V
'0'
0 V


3 V to 5V
High
VOH
VOL
0 V to 0.5 V
Low

Logic gate voltage levels

V_{OH} - Minimum output voltage which will be considered as logic '1'

V_{OL} - Maximum output voltage which will be considered as logic '0'

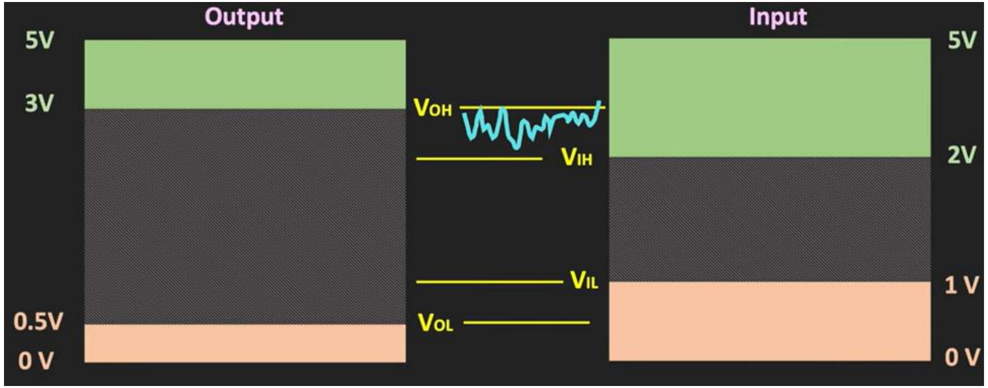
V_{IH} - Minimum Input voltage which will be considered as logic '1'

V_{IL} - Maximum Input voltage which will be considered as logic '0'

Digital Voltage Noise Margin

Consider one digital gate feeding another gate.

- OH - Output High
- OL - Output Low
- IH - Input High
- IL - Input Low
- NM - Noise Margin

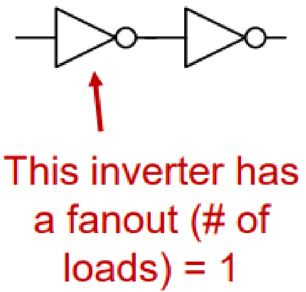


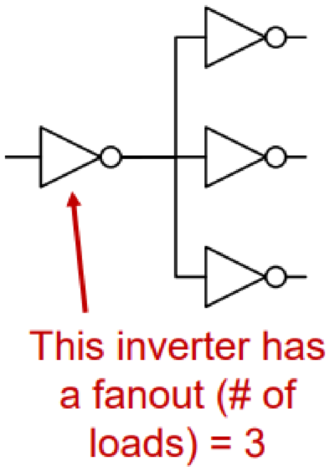
As long as VOH > VIH and VOL < VIL we are in good shape…

Electromagnetic interference & power spikes can cause this to break down

Fanout

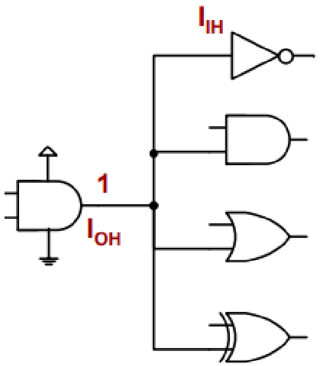
- Fanout refers the number of gates (aka "loads") an output connects to
- As the fanout increases delay increases proportionally
- In addition, if fanout is too high the circuit may stop working
  - Due to current limitations

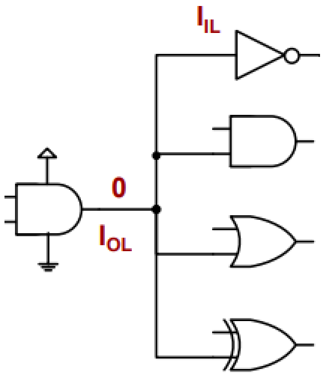




Fanout and Current Limitations

- When a circuit outputs a 'HIGH' ('1') it can only supply (source) so much current = I_{OH}
- When a circuit outputs a 'LOW' ('0') it can only suck up (sink) so much current = I_{OL}
- When a circuit receives a 'HIGH' signal on the input side it may need a certain amount of current to recognize the input as 'HIGH' = I_{IH}
- When a circuit receives a 'LOW' signal on the input side it may need a certain amount of current to recognize the input as 'LOW' = I_{IL}





Example

Dev.	VOH	VIH	VOL	VIL	IOH	IIH	IOL	IIL
A	3.4V	3.3V	0.5V	1.0V	-4 mA	-1 mA	10 mA	2 mA
B	3.2V	3.0V	0.6V	0.7V	-2 mA	-1 mA	6 mA	2 mA

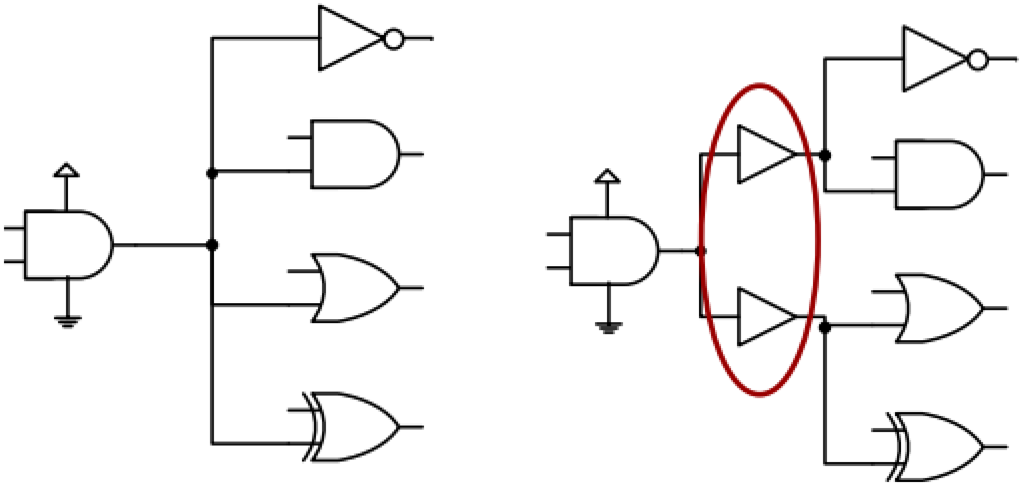
Voltage requirement are compatible!
Device A VOH > Device B VIH
AND Device A VOL < Device B. VIL

Device A's output can drive 4 Device B inputs When outputting '1': - (Dev. A IOH / Dev. B IIH) = (-4 / -1) = 4 When outputting '0': - (Dev. A IOL / Dev. B IIL) = (10 / 2) = 5 Drive capability = min(4, 5) = 4

Consideration

If we attach too many gates to one output it may not be enough to drive those gates

Need to make sure the current requirements and capabilities match



If IOH or IOL is too low we can split the loads by place intermediate buffers

Input Output Current Levels:

INPUT Logic level	Voltage	Current	OUTPUT Logic level	Voltage	Current
0	0.8V max	+/- 10uA max	0	0.45V max	+2mA max
1	2.0V min	+/- 10uA max	1	2.4V min	- 400uA max

8086 and 8088 Device Specifications

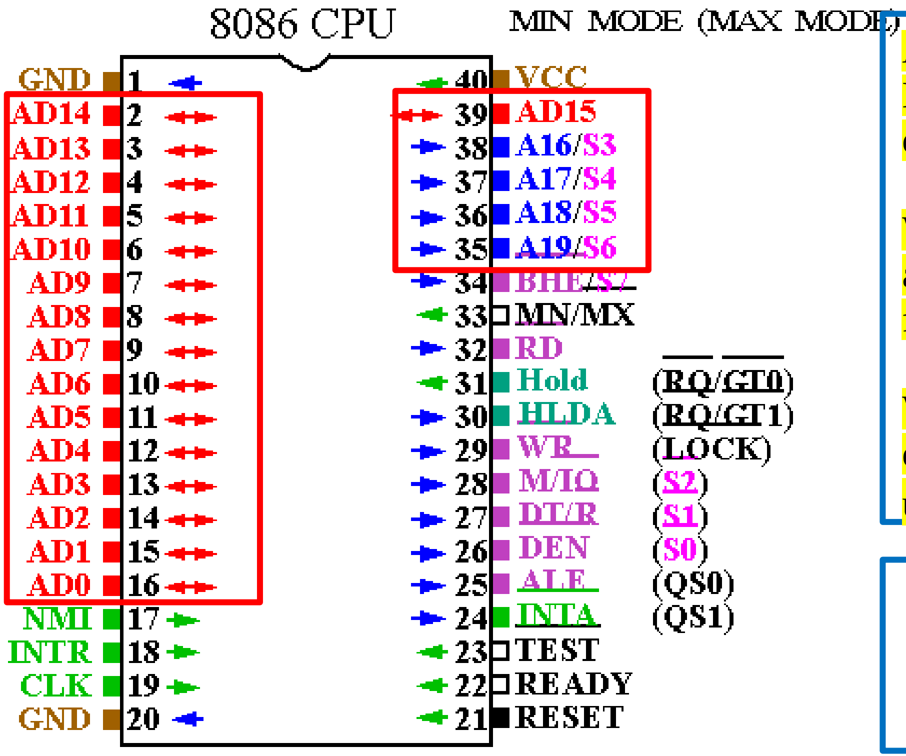
Both are packaged in DIP (Dual In-Line Packages).

- 8086: 16-bit microprocessor with a 16-bit data bus
- 8088: 16-bit microprocessor with an 8-bit data bus.

Both are 5V parts:

- 8086: Draws a maximum supply current of 360mA.
- 8088: Draws a maximum supply current of 340mA.
- 80C86/80C88: CMOS version draws 10mA with temp spec -40 to 225degF.

8086/8088 Pin Out



AD0-AD15(Bidirectional) Address/Data bus : Low order address bus; these are multiplexed with data.

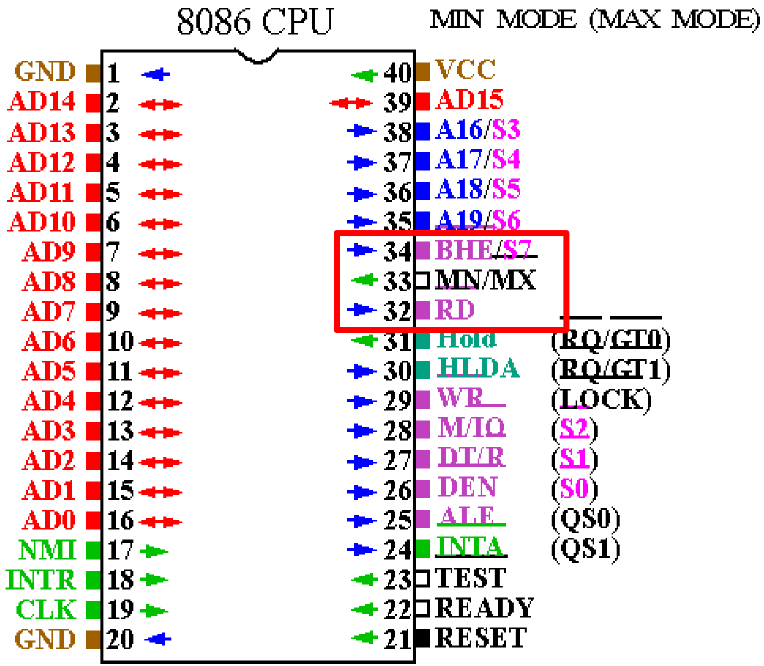
When AD lines are used to transmit memory address the symbol A is used instead of AD, for example A0-A15.

When AD lines are used to transmit data address the symbol D is used instead of AD, for example D0-D15.

A16/S3, A17/S4, A18/S5, A19/S6
High order address bus. These are multiplexed with Status Signals.

Image source https://ece-research.unm.edu/jimp/310/slides/8086_chipset-2.gif

8086/8088 Pin Out



BHE (Active Low)/S7(Output) Bus High Enable/Status

It is used to enable data on to the most significant half of data bus,D8-D15. 8-bit device connected to upper half of the data bus use BHE(Active Low) signal. It is multiplexed with status signal S7.

MN/MX
High Enable/Status

This pin indicates the mode of the processor to operate on.

RD (Read) (Active Low)

The signal is used for read operation.
It is an output signal. It is active when low.

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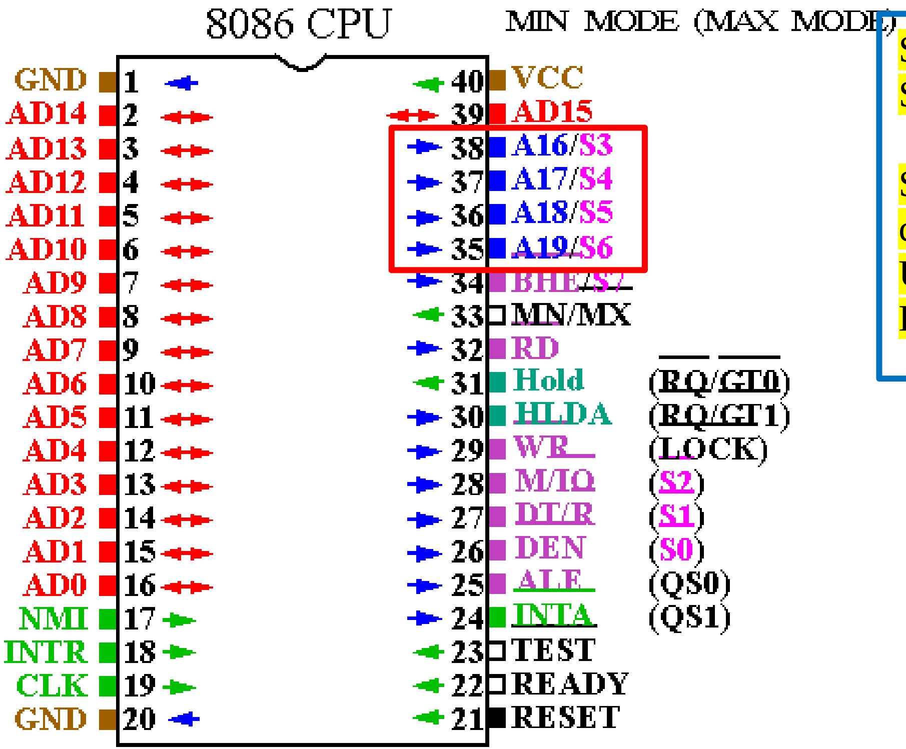
8086/8088 Pin Out



S3, S4, S5, S6 (Status Signals) : S6 is always logic 0
S5 indicates the condition of the IF flag bit.

S4 and S3 show which segment is access during the current bus cycle. These two status bits could be Used to address four separate 1M byte memory Bank by decoding them as A_{21} and A_{20}.

S_4	S_3	*Function*
0	0	Extra segment
0	1	Stack segment
1	0	Code or no segment
1	1	Data segment

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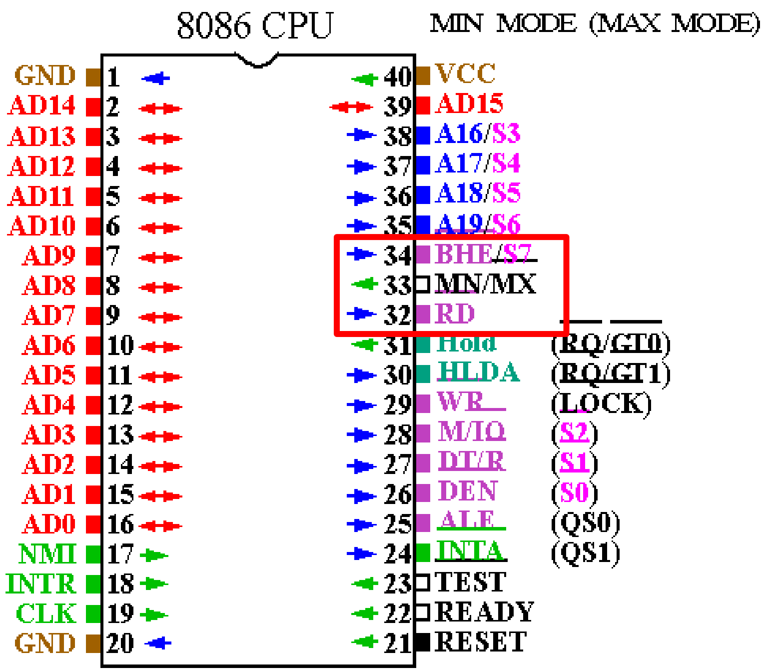

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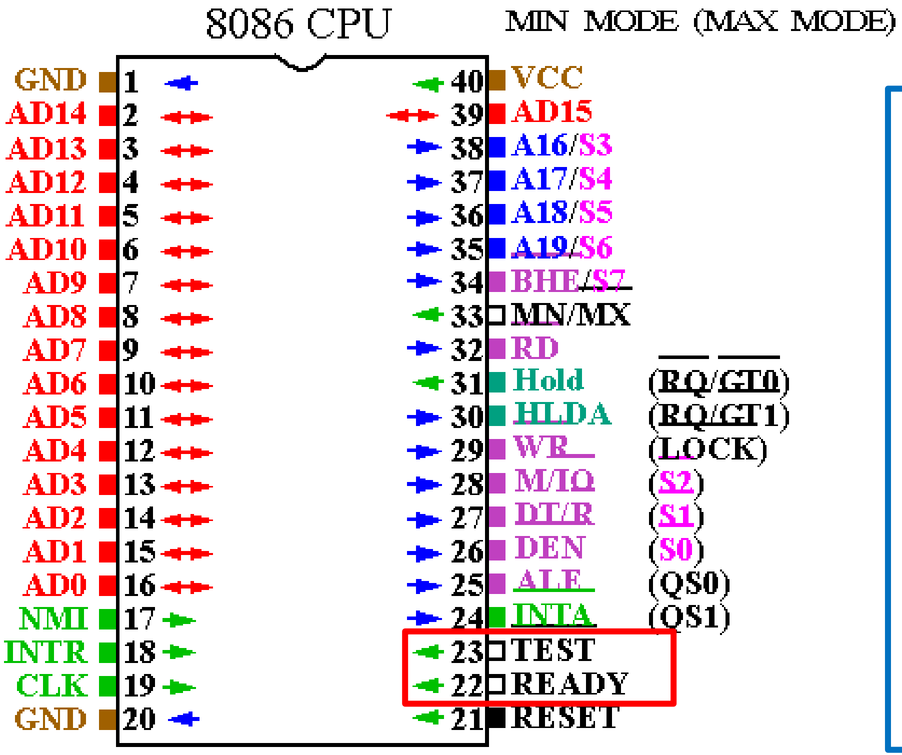

Test

TEST input is tested by the 'WAIT' instruction. 8086 will enter a **wait state** after execution of the **WAIT instruction** and will resume execution only when the **TEST** is made low by an active hardware.

This is used to synchronize an external activity to the processor internal operation.

Ready

This is used to **insert the Wait states** into the timing of the Microprocessor. If this **signal is low** the Microprocessor will enter into **wait states and remains idle**.

Image source https://ece-research.unm.edu/jimp/310/slides/8086_chipset-2.gif